

Diolab – Product Requirements Document (PRD)

AI■Enhanced Cross■Platform SaaS for Diagnostic Centers in India

1. Product Overview

Diolab is a cross■platform SaaS (Web + Mobile) designed for diagnostic and pathology centers in India. It simplifies daily operations such as patient management, billing, reporting, printing, multi■branch control, and communication. AI is embedded wherever possible to reduce manual work, speed up tasks, and improve accuracy.

2. Goals

- Enable billing in under 30 seconds
- Reduce manual data entry using AI assistance
- Improve report accuracy and delivery speed
- Support Indian billing, printers, and WhatsApp workflows
- Work reliably in low■tech environments

3. Platforms

- Web Application (Admin, Billing, Reports)
- Android App (Full functionality)
- iOS App (Full functionality)
- Partial offline support for mobile printing and caching

4. User Roles

- Owner / Super Admin
- Branch Admin
- Front Desk Staff
- Lab Technician
- Collection Agent
- Accountant (View only)

5. Organization, Branding & Branches

- Custom logo upload
- Lab name, address, GST, contact details
- Invoice & report branding
- Multi■branch setup with branch■wise printers and staff

AI Enhancement:

- AI checks logo quality and auto■optimizes for print
- AI suggests footer text and disclaimers

6. Patient Management

- Quick patient registration
- Phone■based search
- Patient history and family grouping

AI Enhancement:

- Auto-fill patient details for returning users
- Detect duplicate patient records

7. Tests & Custom Packages

- Individual tests and categories
- Create custom packages with discounts
- Validity and branch-specific pricing

AI Enhancement:

- AI suggests popular packages based on usage
- Smart price recommendations

8. Billing & Payments

- Fast billing workflow
- Partial payments and dues
- Cash, UPI, Card, Online links
- QR codes on invoices

AI Enhancement:

- Predictive test suggestions during billing
- Anomaly detection in billing amounts

9. Printing & Devices

- Bluetooth & Wi-Fi thermal printer support
- POS bill machines
- Mobile-first printing

AI Enhancement:

- Printer auto-detection
- Auto retry and print error diagnosis

10. Reports & Lab Workflow

- Sample status tracking
- PDF report upload
- Version control

AI Enhancement:

- Auto-flag abnormal values
- AI-generated report summaries for patients

11. Communication

- WhatsApp bill & report sharing
- Email with attachments
- Secure patient portal links

AI Enhancement:

- AI-written patient-friendly messages
- Auto follow-ups for pending reports

12. Inventory (Lightweight)

- Consumable tracking
- Low stock alerts

AI Enhancement:

- Predictive stock usage and reorder suggestions

13. Analytics & Reports

- Daily & monthly reports
- Branch and test-wise revenue

AI Enhancement:

- AI insights on trends and performance
- Smart recommendations for growth

14. Non-Functional Requirements

- Fast performance
- Secure authentication
- Role-based access
- Data encryption and audit logs

15. MVP Scope

Included:

- Patients, tests, packages
- Billing & payments
- Printing
- Report sharing
- Branding & branches

Excluded:

- Insurance integrations
- Hospital EMR integrations

16. Future Roadmap

- Patient mobile app
- WhatsApp Business API automation
- AI voice assistant for front desk
- Corporate health dashboards
- Advanced diagnostics AI